The Chain Rule & Directional Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Multivariable Chain Rule**

Find the directional derivative of $f(x, y) = x^2y - 3y^2$ at the point $P(1, 2)$ in the direction of the vector $\mathbf{v} = \langle 3, 4 \rangle$.

1. Compute the gradient vector *ablaf*: $ablaf(x, y) = \langle 2xy,\ x^2 - 6y \rangle$. At $P(1, 2)$: $ablaf(1, 2) = \langle 2(1)(2),\ 1^2 - 6(2) \rangle = \langle 4,\ -11 \rangle$. 2. Normalize the direction vector $\mathbf{v}$: $||\mathbf{v}|| = \sqrt{3^2 + 4^2} = 5 \Rightarrow \mathbf{u} = \frac{\mathbf{v}}{||\mathbf{v}||} = \left\langle \frac{3}{5},\ \frac{4}{5} \right\rangle$. 3. Compute the directional derivative.

2 **[Foundational] • Target: The Gradient Vector**

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.

We are given $z = f(x, y) = x^2 + y^2$, with $x = 2t$ and $y = 3t$. We need to find $\frac{dz}{dt}$ using the Multivariable Chain Rule: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$. First, find the partial derivatives of z with respect to x and y: $\frac{\partial z}{\partial x} = 2x$ $\frac{\partial z}{\partial y} = 2y$ Next, find the derivatives of x and y with respect to t: $\frac{dx}{dt} = 2$ $\frac{dy}{dt} = 3$ Now, substitute these into.

3 **[Foundational] • Target: Directional Derivatives**

For $f(x, y) = -12x - 35y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

$ablaf = (-12,\ -35)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-12)^2 + (-35)^2 = 37$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

4 **[Foundational] • Target: Direction of Maximum Increase & Steepest Ascent**

For $f(x, y) = -20x + 21y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

$ablaf = (-20, 21)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-20)^2 + (21)^2 = 29$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

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5 [Foundational] • Target: Multivariable Chain Rule

Prove that the 3D vector field $\mathbf{F} = \langle 2xy + z^2,\ x^2 + 2yz,\ 2xz + y^2 \rangle$ is conservative on $\mathbb{R}^3$, construct a potential function f satisfying $ablaf = \mathbf{F}$, and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0, 0)$ to $(1, 2, 3)$.

1. Verify $abla \times \mathbf{F} = \mathbf{0}$: $abla \times \mathbf{F} = \left|\mathbf{i}, \mathbf{j}, \mathbf{k}; \partial_x, \partial_y, \partial_z; 2xy + z^2, x^2 + 2yz, 2xz + y^2\right| = \langle 2y - 2y,\ 2z - 2z,\ 2x - 2x \rangle = \mathbf{0}$. Since $\mathbb{R}^3$ is simply connected, $\mathbf{F}$ is conservative. 2. Reconstruct the potential function:Integrate $f_x = 2xy + z^2 \Rightarrow f(x, y, z) = x^2y + xz^2 + g(y, z)$.Differentiate with respect to y.

6 [Foundational] • Target: The Gradient Vector

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F} = \langle y,\ -x, z \rangle$ along the circular helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \le t \le 2\pi$.

1. Velocity vector: $\mathbf{r}'(t) = \langle -\sin t,\ \cos t,\ 1 \rangle$. 2. Evaluate $\mathbf{F}$ along the curve: With $x = \cos t, y = \sin t, z = t$: $\mathbf{F}(\mathbf{r}(t)) = \langle \sin t,\ -\cos t,\ t \rangle$. 3. Compute dot product: $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = (\sin t)(-\sin t) + (-\cos t)(\cos t) + (t)(1) = -\sin^2 t - \cos^2 t + t = -1 + t$. 4. Integrate.

7 [Foundational] • Target: Directional Derivatives

Consider the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$. What is the value of the vector field at the point $(3,\ -1)$?

To find the value of the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$ at the point $(3,\ -1)$, we substitute $x = 3$ and $y = -1$ into the components of the vector field. The first component is $2x + y = 2(3) + (-1) = 6 - 1 = 5$. The second component is $x - 3y = 3 - 3(-1) = 3 + 3 = 6$. Therefore, the vector field at $(3,\ -1)$ is $\langle 5, 6 \rangle$. Final Answer: $\langle 7, 0 \rangle$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

Maximize $f(x, y) = 5xy$ subject to $x + y = -10$. Find the maximum value.

$g(x, y) = x + y$, $ablag = (1, 1)$. $ablaf = (5y, 5x)$; setting $ablaf = \lambda ablag$ gives $5y = \lambda = 5x \Rightarrow x = y$. With $x + y = -10$: $x = y = -5$, so $f = 125$.

9 [Intermediate] • Target: Multivariable Chain Rule

Find the total mass and the z-coordinate of the center of mass of the upper solid hemisphere $x^2 + y^2 + z^2 \le R^2$, $z \ge 0$, whose density is proportional to the distance from the origin: $\delta(\rho, \phi, \theta) = k\rho$ for constant $k > 0$.

1. Coordinate Setup: In spherical coordinates: $0 \le \rho \le R$, $0 \le \phi \le \frac{\pi}{2}$, $0 \le \theta \le 2\pi$, $dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$. 2. Total Mass M: $M = \iiint_E (k\rho)(\rho^2 \sin\phi) \, d\rho \, d\phi \, d\theta = k\left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi/2} \sin\phi \, d\phi\right)\left(\int_0^R \rho^3 \, d\rho\right)$. $M = k(2\pi)(1)\left(\frac{R^4}{4}\right) = \frac{\pi k R^4}{2}$. 3. Moment with respect to xy-plane M_{xy}: Since.

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10 **[Intermediate]** • **Target: The Gradient Vector**

Compute the moment of inertia I_z about the z-axis of a solid homogeneous right circular cone of base radius R, height h, and uniform total mass M.

1. Cone Geometry & Density: The cone has vertex at the origin and base at $z = h$, so the lateral surface is $r = \frac{R}{h}z \Rightarrow z = \frac{h}{R}r$. Total volume $V = \frac{1}{3}\pi R^2 h$, so density $\delta_0 = \frac{3M}{\pi R^2 h}$. 2. Cylindrical Integral Setup for $I_z = \iiint_E r^2 \delta_0 \, dV$: Bounds: $0 \le \theta \le 2\pi$, $0 \le r \le R$, $\frac{h}{R}r \le z \le h$.

11 **[Intermediate]** • **Target: Directional Derivatives**

Find the volume and the z-coordinate of the centroid $\bar{z}$ of the solid region bounded below by the xy-plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$, assuming uniform density $\delta_0 = 1$.

1. Cylindrical Coordinate Setup: The paraboloid intersects $z = 0$ at $r^2 = 4 \Rightarrow r = 2$. Bounds: $0 \le \theta \le 2\pi$, $0 \le r \le 2$, $0 \le z \le 4 - r^2$, with $dV = r\,dz\,dr\,d\theta$. 2. Volume: $V = \int_0^{2\pi}\int_0^2\int_0^{4-r^2} r\,dz\,dr\,d\theta = 2\pi\int_0^2 r(4-r^2)\,dr = 2\pi[2r^2 - \frac{r^4}{4}]_0^2 = 2\pi(8-4) = 8\pi.$ 3. Moment M_{xy}.

12 **[Intermediate]** • **Target: Direction of Maximum Increase & Steepest Ascent**

Evaluate the Cartesian triple integral $\iiint_B xyz^2 \, dV$ over the rectangular box $B = [0, 1] \times [-1, 2] \times [0, 3]$.

Since the region is a Cartesian box and the integrand is completely separable into univariate functions $f(x, y, z) = x \cdot y \cdot z^2$, Fubini's Theorem allows complete factoring: $\iiint_B xyz^2 \, dV = \left(\int_0^1 x\,dx\right)\left(\int_{-1}^2 y\,dy\right)\left(\int_0^3 z^2\,dz\right)$. Compute each single-variable factor: $\int_0^1 x\,dx = [\frac{x^2}{2}]_0^1 = \frac{1}{2}$. $\int_{-1}^2 y\,dy = [\frac{y^2}{2}]_{-1}^2 = \frac{4-1}{2} = \frac{3}{2}$..

13 **[Intermediate]** • **Target: Multivariable Chain Rule**

Evaluate the double integral $\iint_R x\cos(x + y)\,dA$ over the rectangular region $R = [0, \pi] \times [0, \pi/2]$.

1. Set up the iterated integral: $\int_0^{\pi}\int_0^{\pi/2} x\cos(x + y)\,dy\,dx$. 2. Integrate with respect to y: $\int_0^{\pi/2}\cos(x + y)\,dy = [\sin(x + y)]_0^{\pi/2} = \sin(x + \pi/2) - \sin x = \cos x - \sin x.$ 3. Integrate with respect to x using integration by parts: $\int_0^{\pi} x(\cos x - \sin x)\,dx = \int_0^{\pi} x\cos x\,dx - \int_0^{\pi} x\sin x\,dx$. $\int_0^{\pi} x\cos x\,dx = [x\sin x]_0^{\pi} - \int_0^{\pi}\sin x\,dx = 0 - [-\cos x]_0^{\pi} = -(1 - (-1)) = -2$..

14 **[Intermediate]** • **Target: The Gradient Vector**

Classify the critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 + y^3 - 3xy$ using the Second Derivative Test and Hessian matrix.

1. Compute the second partial derivatives: $f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = -3.$ The Hessian determinant discriminant is: $D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = (6x)(6y) - (-3)^2 = 36xy - 9.$ 2. Evaluate at $(0, 0)$: $D(0, 0) = 36(0) - 9 = -9 < 0.$ Since $D < 0$, $(0, 0)$ is a saddle point. 3. Evaluate at $(1, 1)$: $D(1, 1) = 36(1)(1) - 9 = 27 > 0, \quad f_{xx}(1, 1) = 6(1) = 6 > 0.$ Since $D > 0$ and $f_{xx} > 0$, $(1, 1)$ is a local minimum with value $f(1, 1) = 1 + 1 - 3 = -1$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Directional Derivatives • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives**

Consider the solid region E bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 16$. The volume of E is given by the triple integral $V = \iiint_E dV$. If E is expressed in spherical coordinates, the integral becomes $\int_{\alpha}^{\beta} \int_{\gamma}^{\delta} \int_{\rho_1}^{\rho_2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$. Determine the value of α, the lower limit of integration for θ.

The region E is a spherical cone. The upper bound is a sphere of radius 4 centered at the origin, so ρ ranges from 0 to 4. The lower bound is the cone $z = \sqrt{x^2 + y^2}$. In spherical coordinates, $z = \rho\cos\phi$ and $\sqrt{x^2 + y^2} = \rho\sin\phi$. Thus, the cone equation becomes $\rho\cos\phi = \rho\sin\phi$. For $\rho eq 0$, this simplifies to $\cos\phi = \sin\phi$, which means $\tan\phi = 1$. In the range $0 \le \phi \le \pi$, this occurs at $\phi = \frac{\pi}{4}$. Therefore.

16 **[Challenging] • Target: Direction of Maximum Increase & Steepest Ascent • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers**

Find the surface area of the paraboloid cap $z = x^2 + y^2$ that lies below the horizontal plane $z = 4$.

1. Surface Area Differential for a Graph $z = g(x, y)$: $g_x = 2x, \quad g_y = 2y \Rightarrow dS = \sqrt{1 + (2x)^2 + (2y)^2}\, dA = \sqrt{1 + 4(x^2 + y^2)}\, dA.$ 2. Projection onto xy-plane: The intersection $x^2 + y^2 = 4$ is a disk D of radius 2. In polar coordinates: $0 \le r \le 2$, $0 \le \theta \le 2\pi$, and $dA = r\,dr\,d\theta$. 3. Evaluate the surface integral.

17 **[Challenging] • Target: Multivariable Chain Rule • Cross-Lesson: 4.01 Double Integrals**

Prove that the limit $\lim_{(x, y)\to(0, 0)} \frac{x^2 y}{x^4 + y^2}$ does not exist by using the Two-Path Test, comparing straight-line approaches with parabolic approaches.

1. Approach along linear paths $y = mx$: $\lim_{x\to0} \frac{x^2(mx)}{x^4 + (mx)^2} = \lim_{x\to0} \frac{mx^3}{x^2(x^2 + m^2)} = \lim_{x\to0} \frac{mx}{x^2 + m^2} = \frac{0}{m^2} = 0 \quad (\text{for } meq0).$ Along $y = 0$ (the x-axis), $\lim_{x\to0} \frac{0}{x^4} = 0$. Along all straight lines through the origin, the limit is 0. 2. Approach along parabolic paths $y = kx^2$.

18 **[Challenging] • Target: The Gradient Vector • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives**

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

To find the critical points of the function $f(x, y) = x^3 + y^3 - 3axy$, we first compute the first partial derivatives and set them to zero: $f_x = 3x^2 - 3ay$ $f_y = 3y^2 - 3ax$ Setting these to zero, we get the system of equations: 1) $3x^2 - 3ay = 0$ => $x^2 = ay$ 2) $3y^2 - 3ax = 0$ => $y^2 = ax$ We are given that $(x_0, y_0) = (2, 2)$ is a critical point. Substituting these values into the equations: From equation 1: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$. From equation 2: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$. Bo.

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19 [Challenging] • Target: Directional Derivatives • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the region R in the xy-plane defined by $0 \le x \le 1$ and $0 \le y \le x^2$. Evaluate the double integral $\iint_R y\,dA$.

The double integral can be evaluated using Fubini's Theorem as an iterated integral. The region of integration is described by $0 \le x \le 1$ and $0 \le y \le x^2$. Therefore, the integral is set up as: $\int_0^1 \int_0^{x^2} y\,dy\,dx$ First, we integrate with respect to y: $\int_0^{x^2} y\,dy = \left[\frac{y^2}{2}\right]_0^{x^2} = \frac{(x^2)^2}{2} - \frac{0^2}{2} = \frac{x^4}{2}$ Now, we integrate the result with respect to x.

20 [Challenging] • Target: Direction of Maximum Increase & Steepest Ascent • Cross-Lesson: 4.01 Double Integrals

A solid region E is defined by the inequalities $x^2 + y^2 \le 16$, $0 \le z \le 5$. The volume of this region can be computed using a triple integral. In cylindrical coordinates, the region E is described by $0 \le r \le R$, $0 \le \theta \le 2\pi$, and $0 \le z \le H$. What is the value of R?

The given region is a cylinder with radius R_{cyl} and height H_{cyl}. The inequality $x^2 + y^2 \le 16$ describes a circle of radius 4 centered at the origin in the xy-plane. When converting to cylindrical coordinates, $x^2 + y^2$ becomes r^2. Thus, the inequality $r^2 \le 16$ implies $r \le 4$. Therefore, in cylindrical coordinates, the radial component of the region is described by $0 \le r \le 4$. The question states that the region is described by $0 \le r \le R$, so $R = 4$. The inequality.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multivariable Chain Rule

If $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$, what is $\frac{dz}{dt}$?

a. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ ✓

b. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dy}{dt} + \frac{\partial z}{\partial y}\frac{dx}{dt}$

c. $\frac{dz}{dt} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\frac{dz}{dt} = \frac{dx}{dt}\frac{dy}{dt}$

By the multivariable chain rule, rates of change sum across all intermediate paths: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$.

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22 [Foundational] • Target: Multivariable Chain Rule

Given $z = x^2y$, $x = t^3$, and $y = t^2$, find $\frac{dz}{dt}$ at $t = 1$.

a. 8 ✓

b. 6

c. 5

d. 12

$z = (t^3)^2(t^2) = t^6 \cdot t^2 = t^8$. Therefore $\frac{dz}{dt} = 8t^7$. At $t = 1$, $\frac{dz}{dt} = 8(1)^7 = 8$.

23 [Foundational] • Target: Multivariable Chain Rule

If $z = f(x, y)$ with $x = r\cos\theta$ and $y = r\sin\theta$, what is $\frac{\partial z}{\partial r}$?

a. $\frac{\partial z}{\partial x}\cos\theta + \frac{\partial z}{\partial y}\sin\theta$ ✓

b. $-r\frac{\partial z}{\partial x}\sin\theta + r\frac{\partial z}{\partial y}\cos\theta$

c. $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\cos\theta + \sin\theta$

$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial z}{\partial x}\cos\theta + \frac{\partial z}{\partial y}\sin\theta$.

24 [Foundational] • Target: The Gradient Vector

What is the gradient vector $\nabla f(x, y)$ of a scalar function $f(x, y)$?

a. $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$ ✓

b. $\nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$

c. $\nabla f = \left[\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}\right]$

d. $\nabla f = \sqrt{f_x^2 + f_y^2}$

The gradient of a multivariable function is the vector consisting of its first-order partial derivatives: $\nabla f = [f_x, f_y]$.

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25 [Foundational] • Target: The Gradient Vector

Compute the gradient $\nabla f(2, -1)$ for $f(x, y) = 3x^2y + 2y^2$.

a. $[-12, 8]$ ✓

b. $[12, -8]$

c. $[-12, -4]$

d. $[6, 4]$

$f_x = 6xy \Rightarrow f_x(2, -1) = 6(2)(-1) = -12$. $f_y = 3x^2 + 4y \Rightarrow f_y(2, -1) = 3(4) + 4(-1) = 12 - 4 = 8$. Gradient is $[-12, 8]$.

26 [Foundational] • Target: The Gradient Vector

What is the geometric relationship between the gradient $\nabla f(x_0, y_0)$ and the level curve $f(x, y) = c$ passing through (x_0, y_0)?

a. $\nabla f(x_0, y_0)$ **is perpendicular (orthogonal) to the tangent line of the level curve** ✓

b. $\nabla f(x_0, y_0)$ is parallel to the tangent line

c. $\nabla f(x_0, y_0)$ has zero dot product with the normal vector

d. $\nabla f(x_0, y_0)$ points along the curve

The gradient vector at any point is always perpendicular to the level curve (or level surface) passing through that point.

27 [Foundational] • Target: Directional Derivatives

What is the formula for the directional derivative $D_{\mathbf{u}}f$ in the direction of a unit vector $\mathbf{u}$?

a. $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ ✓

b. $D_{\mathbf{u}}f = \nabla f \times \mathbf{u}$

c. $D_{\mathbf{u}}f = ||\nabla f|| ||\mathbf{u}||$

d. $D_{\mathbf{u}}f = \frac{\nabla f \cdot \mathbf{u}}{||\nabla f||}$

The rate of change of f in the direction of unit vector $\mathbf{u}$ is the dot product of the gradient with the unit vector: $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

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28 [Foundational] • Target: Directional Derivatives

If $\nabla f(1,2) = [4, -3]$ and $\mathbf{u} = \left[\frac{3}{5}, \frac{4}{5}\right]$, find $D_{\mathbf{u}} f(1,2)$.

a. 0 ✓

b. 5

c. $\frac{24}{5}$

d. -1

$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = (4)\left(\frac{3}{5}\right) + (-3)\left(\frac{4}{5}\right) = \frac{12}{5} - \frac{12}{5} = 0.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Directional Derivatives

Why must $\mathbf{u}$ be normalized to a unit vector before computing $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$?

a. **To ensure the derivative measures rate of change per unit of geometric distance** ✓

b. Because dot products cannot be computed with non-unit vectors

c. To avoid division by zero

d. Because the gradient has length 1

Normalizing $\mathbf{u}$ ensures that the directional derivative accurately reflects the instantaneous rate of change per unit distance along that direction.

30 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

In which direction does a differentiable function f increase most rapidly at a point?

a. **In the direction of the gradient vector** ∇f ✓

b. In the direction opposite to ∇f (i.e. $-\nabla f$)

c. Orthogonal to the gradient vector

d. Along the tangent to the level curve

Since $D_{\mathbf{u}} f = ||\nabla f|| \cos\theta$, the maximum rate occurs when $\cos\theta = 1$ ($\theta = 0$), pointing in the exact direction of ∇f.

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31 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

What is the maximum value of the directional derivative $D_{\mathbf{u}}f$ at a point?

a. $||\nabla f||$ ✓

b. $||\nabla f||^2$

c. 0

d. 1

When $\mathbf{u}$ points in the direction of ∇f, $D_{\mathbf{u}}f = \nabla f \cdot \frac{\nabla f}{||\nabla f||} = ||\nabla f||$.

32 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

In which direction is the rate of change of f equal to zero?

a. **In any direction tangent to the level curve (orthogonal to ∇f)** ✓

b. In the direction of $-\nabla f$

c. In the direction of ∇f

d. Nowhere, a differentiable function always changes

When $\mathbf{u}$ is orthogonal to ∇f, $\cos\theta = 0$, so $D_{\mathbf{u}}f = 0$, which occurs along tangent directions to level curves.

33 [Intermediate] • Target: Multivariable Chain Rule

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

a. 6

b. **12** ✓

c. -6

d. 15

To find $f_x(2, -3)$, we first compute the partial derivative of $f(x, y)$ with respect to x. Treating y as a constant, we get: $f_x(x, y) = \frac{\partial}{\partial x}(3x^2 - 5y) = 6x$. Now, we evaluate f_x at the point $(2, -3)$: $f_x(2, -3) = 6(2) = 12$.

34 [Intermediate] • Target: The Gradient Vector

For $f(x, y) = -9x - 40y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. **41** ✓

b. 49

c. 40

d. 42

$ablaf = (-9, -40)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-9)^2 + (-40)^2 = 41$.

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35 [Intermediate] • Target: Directional Derivatives

For $f(x, y) = 20x - 21y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 41

b. **29** ✓

c. 28

d. 30

$ablaf = (20, -21)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (20)^2 + (-21)^2 = 29$.

36 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

For $f(x, y) = -15x - 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 23

b. 16

c. 18

d. **17** ✓

$ablaf = (-15, -8)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-15)^2 + (-8)^2 = 17$.

37 [Intermediate] • Target: Multivariable Chain Rule

For $f(x, y) = -7x - 24y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 31

b. **25** ✓

c. 24

d. 26

$ablaf = (-7, -24)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-7)^2 + (-24)^2 = 25$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The Gradient Vector • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = -24x - 7y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 31

b. 24

c. 26

d. **25** ✓

$ablaf = (-24, -7)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-24)^2 + (-7)^2 = 25$.

39 [Challenging] • Target: Directional Derivatives • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = -21x + 20y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 41

b. **29** ✓

c. 28

d. 30

$ablaf = (-21, 20)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-21)^2 + (20)^2 = 29$.

40 [Challenging] • Target: Direction of Maximum Increase & Steepest Ascent • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 8x - 15y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 23

b. 16

c. 18

d. **17** ✓

$ablaf = (8, -15)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (8)^2 + (-15)^2 = 17$.

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41 [Challenging] • Target: Multivariable Chain Rule • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 24x - 7y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 31

b. 24

c. 26

d. **25** ✓

$ablaf = (24, -7)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (24)^2 + (-7)^2 = 25$.

42 [Challenging] • Target: The Gradient Vector • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = -15x + 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 23

b. 16

c. 18

d. **17** ✓

$ablaf = (-15, 8)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-15)^2 + (8)^2 = 17$.

43 [Challenging] • Target: Directional Derivatives • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = -6x - 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. **10** ✓

b. 14

c. 9

d. 11

$ablaf = (-6, -8)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-6)^2 + (-8)^2 = 10$.

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44 [Challenging] • Target: Direction of Maximum Increase & Steepest Ascent • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 12x - 5y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. **13** ✓

b. 17

c. 12

d. 14

$ablaf = (12, -5)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (12)^2 + (-5)^2 = 13$.

45 [Challenging] • Target: Multivariable Chain Rule • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = -9x + 40y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 49

b. 40

c. **41** ✓

d. 42

$ablaf = (-9, 40)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-9)^2 + (40)^2 = 41$.

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